

ing to batting crease in the following relationship:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Speed} = \frac{18}{\text{Time}} \times 3.6 \text{ km/hour}$$

The sensor can be made by pasting two reflective mirrors on wooden bases. The mirrors must be placed face to

face on the sides of the bowling crease as shown in Fig. 2. Fix a laser torch on top of one mirror. The laser beam must make a small angle after falling on the mirror placed in front of it so that the beam reflects back and forth between the mirrors until it reaches the LDR placed at the bottom of the mirror. This way the laser beam forms a light barrier. When the ball interrupts the beam, it produces a pulse. Place another such

sensor at the batting side.

Assemble the circuit on a general-purpose PCB and enclose in a small cabinet. Wire the clock buzzer to the circuit and keep this alarm watch at a convenient place. In place of a 6V battery, you can use a 6V adaptor for powering the circuit. Mount each of the LEDs and the pushbutton on the front panel, and connect them up to the board. ●

DIGITAL THERMOMETER

■ RAJ K. GORKHALI

This digital thermometer can measure temperatures up to 150°C with an accuracy of ±1°C. The temperature is read on a 1V full scale-deflection (FSD) moving-coil voltmeter or digital voltmeter.

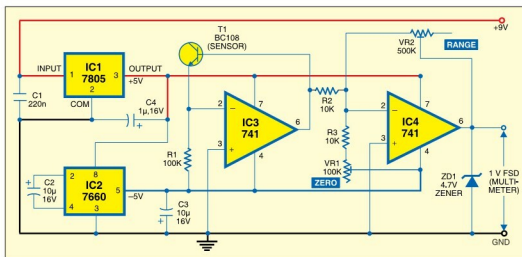
Operational amplifier IC 741 (IC3) provides a constant flow of current

through the base-emitter junction of npn transistor BC108 (T1). The voltage across the base-emitter junction of the transistor is proportional to its temperature. The transistor used this way makes a low-cost sensor. You can use silicon diode instead of transistor. The small variation in voltage across the base-emitter junction is amplified by second operational amplifier (IC4),

before the temperature is displayed on the meter. Preset VR1 is used to set the zero-reading on the meter and preset VR2 is used to set the range of temperature measurement.

Operational amplifiers IC3 and IC4 operate off regulated ±5V power supply, which is derived from 3-terminal positive voltage regulator IC 7805 (IC1) and negative low-dropout regulator IC 7660 (IC2). The entire circuit works off a 9V battery.

Assemble the circuit on a general-purpose PCB and enclose in a small plastic box. Calibrate the thermometer using presets VR1 and VR2. After calibration, keep the box in the vicinity of the object whose temperature is to be measured. ●



NIGHT VISION ENHANCER

■ T.K. HAREENDRAN

Here is a simple green LED flashlight built around 555 timer IC (IC1) and powered

from a 3V battery pack.

Green LED based flashlights are widely used by hunters and others in similar activities to light up their way and alert other hunters of their pres-

ence. Green light does not interfere with the hunt, as animals are blind or

